

Concept Note

Follow-On Peer-to-Peer Technical Workshop

Piloting Interoperable Hydrometeorological Data Systems for Multi-Hazard Early Warning Systems

Background

Strengthening the operational use of water and weather data is central to the development of early warning systems. Across East Africa and beyond, increasing extreme weather, flood and drought risk, and growing pressure on water resources are placing new demands on early warning systems and decision-making institutions.

While significant progress has been made in advancing weather and water policy frameworks, many countries continue to face practical challenges in accessing, integrating, and applying hydrological and meteorological data for operational use. Fragmented information systems, limited interoperability, and uneven institutional capacity constrain the timely production and dissemination of early warning information. Addressing these barriers requires strengthening the accessibility, exchange, and operational use of Earth system data, as identified in the World Meteorological Organization's (WMO) Unified Data Policy, which promotes shareable and authoritative data and interoperable practices that enable timely and equitable access early warning systems.

On April 28–29, 2026, the Center for Geospatial Solutions (CGS) at the Lincoln Institute of Land Policy, the World Meteorological Organization (WMO), and the GEOGLOWS Initiative, with technical support from Utah State University, jointly convened a peer-to-peer technical workshop in Entebbe that brought together the national hydrological and meteorological services of Ethiopia, Kenya, Rwanda, and Uganda. The workshop focused on open-source and interoperable hydrometeorological data systems for multi-hazard early warning systems (MHEWS), highlighting the need to improve forecasting, data management, and operational data exchange while identifying common challenges related to interoperability, data sharing, staffing, technical capacity, and long-term sustainability.

Participants expressed particular interest in practical implementation support, including opportunities to pilot open-source interoperable workflows using technologies such as HydroServer2 and wis2box. They also emphasized the value of peer learning and requested continued technical engagement beyond the initial workshop.



Goal

Support participating National Meteorological and Hydrological Services (NMHSs) in defining and implementing a practical pathway toward country ownership of interoperable hydrometeorological data flows.

Approach

Building on the outcomes of the Entebbe workshop, the proposed workplan follows a phased approach that moves participating countries from pilot testing toward sustained operational adoption of interoperable hydrometeorological data systems.

Phase 1: Pre-Workshop Follow-Up Consultations with the Center for Geospatial Solutions (CGS) (June–August 2026)

CGS will coordinate with heads of hydrological and meteorological institutions to appoint a focal point from each institution that participated in the April 2026 Entebbe workshop to coordinate the activities aiming to:

- Better understand operational priorities and implementation constraints.
- Assess existing technical infrastructure and capacity.
- Identify opportunities to build on existing national and regional initiatives around open-source software.
- Define a practical pilot use case for testing during the workshop (Phase 2).
- Assist in refining the agenda and prioritizing topics for workshop (Phase 2).

The results of these consultations will inform the agenda and technical focus of the follow-up workshop.

Phase 2: Follow-Up Peer-to-Peer Technical Workshop: Piloting Interoperable Hydrometeorological Data Systems for Multi-Hazard Early Warning Systems (September 14–17, 2026, Addis Ababa)

The technical workshop will bring together hydrological and meteorological services, WMO experts, software developers, IT specialists, and implementation partners. Moving beyond demonstrations, participants will execute hands-on use cases including:

- Configure and test interoperable systems: Explore the installation, configuration, and use of open-source software, including data management, metadata, quality control, and system interoperability.
- Validate operational workflows: Scoping possible end-to-end data flows that integrate existing national systems with open-source platforms, supporting the publication, exchange, and operational use of hydrometeorological data for forecasting and early warning.
- Identify software enhancements, bug fixes, and interoperability improvements emerging from hands-on testing with participating countries.
- Develop implementation Roadmap: Developing country-specific roadmaps outlining the technical, operational, and institutional steps required to deploy, operate, maintain, and scale the piloted open-source solutions, while identifying opportunities to align with and contribute to ongoing regional initiatives, coordination mechanisms, and knowledge-sharing activities.
- Plan for institutional sustainability: For institutions interested in moving forward following the pilot, identify the institutional arrangements needed for long-term implementation and scaling, including ownership and governance, organizational roles and responsibilities, and capacity development.

Participants will work collaboratively to test use cases and develop the roadmap while technical partners identify software enhancements and support requirements.

Phase 3: Demand-Driven Implementation Support and Transition to Country Ownership (September 2026–December 2026)

Following the workshop, participating NMHSs that wish to pursue operational implementation of HydroServer2 or wis2box will be invited to submit a request for follow-on technical support to CGS and WMO. For six months following the workshop, CGS, WMO, and collaborating partners will support continued peer exchange through periodic technical follow-up sessions, targeted training activities, and demand-driven learning materials identified by participating experts and institutions. Throughout this period, we will actively foster peer-to-peer collaboration and knowledge exchange among participating NMHSs, with the goal of enabling countries to increasingly support one another and establishing a sustainable community of practice that can continue beyond the project's direct involvement. These efforts will help participating institutions move from planning toward operational implementation while continuing to share lessons about learned and practical experiences across the region.

Expected Results

- Shared understanding of progress, lessons learned, and challenges since the April 2026 peer-to-peer technical workshop in Entebbe Uganda.
- Improved understanding of practical approaches to applying integrated weather and water data in operational early warning and risk-informed decision-making through the piloting of interoperable and open-source software solutions.
- Country-specific implementation roadmaps outlining the technical, operational, and institutional steps required to deploy, operate, maintain, and sustain the piloted interoperable and open-source software solutions, should countries choose to implement them.
- Validated pilot use cases and technical requirements to guide software integration, future development, and operational adoption within participating NMHSs.
- Prioritized software enhancements, bug fixes, and interoperability improvements identified through hands-on testing with participating countries.
- Workshop materials and reference resources shared with participants following the event to support continued application, implementation, and knowledge sharing.
- Strengthened collaboration among NMHSs, WMO, software developers, regional organizations, and implementation partners, supported by an active regional peer-to-peer community of practice for continued knowledge exchange and technical support.
- A practical pathway toward country ownership of interoperable hydrometeorological data flows, enabling participating institutions to progressively implement, operate, and sustain these systems with decreasing reliance on external technical support.

Participants (estimated 45 total)

- Senior officials and decision-makers from the National Hydrological and Meteorological Services of Ethiopia, Kenya, Rwanda, and Uganda
- Operational hydrologists, meteorologists, forecasters, and IT specialists from the National Hydrological and Meteorological Services of Ethiopia, Kenya, Rwanda, and Uganda
- Regional and international technical partners

Program Structure (September 14–17, 2026, Addis Ababa)

Day 1 will review the key lessons from the Entebbe workshop together with the findings of the pre-workshop consultations, highlighting common implementation priorities, country-specific pilot activities, and shared technical challenges across participating NMHSs. Guest contributions

from Okoth Kabaka (ICPAC) and Mohammed Ali (WMO) will provide additional regional and technical perspectives to inform the discussions. Together, these sessions will establish a common foundation for the hands-on technical implementation and roadmap development activities that follow.

Day 2 will provide parallel hands-on training on HydroServer2 and wis2box. Participants will work with software developers to configure the platforms, ingest and manage data, publish observations, and explore operational deployment considerations using country-relevant datasets and workflows. The training will also demonstrate how these platforms support the publication, discovery, and exchange of hydrometeorological data through WIS2 and the WMO Hydrological Observing System (WHOS), where appropriate.

Day 3 will focus on integrating interoperable open-source software with existing national systems and operational workflows. Through practical exercises, participants will develop end-to-end data flows, refine country pilot activities, and identify software enhancements, technical requirements, and implementation considerations. Guest presentations from NBI, CIAT, and WMO's Regional Office for Africa will bring complementary regional perspectives and implementation experience to inform the technical discussions and country pilots.

Day 4 will focus on developing country-specific implementation roadmaps that outline the technical, operational, and institutional requirements for deploying, operating, and maintaining the piloted software solutions, should participating institutions decide to adopt them following the workshop. The roadmaps will identify the technical and institutional steps needed to operationalize interoperability through WIS2 and WHOS, as appropriate, while also identifying opportunities for continued regional collaboration and peer-to-peer support.

Roles and Responsibilities

Center for Geospatial Solutions (CGS)

Coordinate the workshop, co-facilitate sessions, and co-deliver introductory exercises on interoperable hydrological and meteorological data workflows. Document key discussion points and agreed follow-up considerations.

World Meteorological Organization (WMO)

Provide strategic leadership on hydrometeorological data interoperability, including alignment with WIS2, WHOS, and the Early Warnings for All (EW4All) initiative. Support engagement with national meteorological and hydrological services and coordinate with partners to ensure the workshop reflects WMO Member priorities and advances operational implementation.

Participant materials (highly recommended)

- Personal laptop

Documentation and Reporting

Proceedings will be documented by the organizing partners. A summary report will capture key discussions, priority actions, and recommended next steps. Workshop materials and reference resources will be shared with participants following the event to support continued application.

Partnership and Alignment

The workshop is designed to complement ongoing national, regional, and global initiatives related to hydrological services and early warning systems, including through demonstration of public–private partnership approaches that support data integration and service delivery. Coordination with partner organizations will help ensure coherence with broader capacity development and implementation efforts.

Resourcing

The workshop benefits from in-kind technical contributions from the World Meteorological Organization. Financial support for the organization of the event and participant participation is provided by the Center for Geospatial Solutions at the Lincoln Institute of Land Policy through a project implemented with the National Oceanic and Atmospheric Administration (NOAA) International Office of the National Weather Service (NWS) and funded by the United States Government.

Workshop Agenda

Day 1: September 14	
Time	Session

08:00 – 09:00	• Welcome • Sign in • Coffee
09:00 – 10:40	• Opening remarks Dr. Fetene Teshome, Ethiopian Meteorological Institute (EMI), Mr. Habtemaryam Tilahun, Hydrology and Basin Information, Ms. Anasthasia Agyemang, US Embassy in Ethiopia, Mr. Washington Otieno, Regional Office for Africa (RAF) World Meteorological Organization (WMO), Dr. Caroline Staub, Center for Geospatial Solutions (CGS) • Partner and delegation introductions • Key outcomes and lessons from the Entebbe workshop and follow-up consultations • Workshop agenda and objectives
10:40 – 11:00	Break
11:00 – 12:30	• Session 1: Multi-Hazard Early Warning Systems (MHEWS): Objectives, Workflows, and Operational Challenges Mr. Zablon Shilenje, Mr. Washington Otieno ○ Objectives ○ Common bottlenecks during dissemination. ○ Why open standards, open sources, and interoperability matter ○ Guest speaker: Dr. Okoth Kabaka (ICPAC)
12:30 – 13:45	Lunch

13:45 – 15:15	- Session 2: Group activity on hydrometeorological data flow. Ms. Emily DuBois, Dr. Caroline Staub - Identify (1) map where participants work within the data system, (2) where they see the greatest opportunity for improvement, and (3) where they experience the greatest challenges or friction - Reflect as a group on emerging patterns, challenge & opportunity clusters
15:15 – 15:35	Break
15:35 – 17:00	- Session 3: Guest presentation and preparation for Day 2 Guest speaker: Mr. Mohammed Ali (WMO) <i>Set up for Day 2</i>

Day 2: September 15 – wis2box and HydroServer Training Agenda for National Meteorological and Hydrological Services		
Time	Session	
08:00 – 09:00	- Welcome - Sign in - Coffee	
09:00 – 10:40	Session 4a: Introduction to HydroServer: Architecture,	Session 4b: Introduction to WIS2 and wis2box Mr. Peter Mutai

	Data Models & Deployment Options Dr. Sara Alonso Vicario ○ HydroServer core concepts, data model, and OGC standards ○ Deployment options (local, cloud, hybrid) ○ Operational considerations	○ WIS2 architecture and components ○ Data discovery, metadata, and messaging concepts ○ Introduction to wis2box and its role within WIS2
10:40 – 11:00	Break	
11:00 – 12:30	• Session 5a: Getting Started with HydroServer: Setup, Data Management, and Observations Dr. Sara Alonso Vicario ○ Creating workspaces, monitoring sites, and data streams ○ Understanding user roles and data privacy ○ Loading Historical Observations	• Session 5b: Installing and configuring wis2box Mr. Peter Mutai ○ wis2box architecture ○ Docker services and system components ○ wis2box Web Application and API ○ Registering your WIS2 Node
12:30 – 13:45	Lunch	
13:45 – 15:15	• Session 6a: Working with HydroServer: Automated Data Ingestion, Visualization, and Quality Control Dr. Sara Alonso Vicario	• Session 6b: Publishing Data through WIS2 Mr. Peter Mutai

	○ Automated Data Ingestion ○ Data Visualization ○ Data Quality Control	○ Visualizing observations, exploring, and editing metadata ○ Forecasting and Quality control ○ Publishing different data types
15:15 – 15:35	Break	
15:35 – 17:30	• Session 7a: Querying Data and Automating Data Ingestion in HydroServer Dr. Sara Alonso Vicario ○ Querying and exporting data and metadata from observations ○ Other automated data ingestion options	• Session 7b: Accessing WIS2 Data and Operational Transition Mr. Peter Mutai ○ GTS Bridge ○ Discovering datasets from the Global Catalogue and ○ Downloading data using WIS2

Day 3: September 16	
Time	Session
08:00 – 09:00	• Welcome • Sign in • Coffee

09:00 – 10:40	- Session 8: HydroServer-WIS2Box Data Workflows and Data Readiness for the Pilot Dr. Sara Alonso Vicario, Mr. Washington Otieno - Overview of WMO Hydrological Observing System (WHOS) for hydrological data exchange and its relationship with WIS2 - Identify priority datasets (real- time and historical) - Assessing WIS2Box and HydroServer integration options, including API availability, security requirements, and strategies to address implementation challenges. Guest speaker: Mr. Alain Gerard Amahoro (Nile Basin Initiative)
10:40 – 11:00	Break
11:00 – 12:30	- Session 9: Working with Existing and New Data Management Systems Dr. Sara Alonso Vicario, Mr. Washington Otieno - Explore existing databases and data management systems countries are using - Explore possible data connections between data management systems, with some examples for HydroServer and Aquarius and wis2box and Climsoft Guest speaker: Dr. Jemal S. Ahmed (CIAT)
12:30 – 13:45	Lunch
13:45 – 15:15	- Session 10: Interactive Exercise: Refining Pilots to Improve Data Management Workflows for MHEWS Ms. Emily DuBois, Dr. Caroline Staub - Group exercise mapping data flows from observation to delivery of information products to users

○ Bridging the gap between National Hydrological and Meteorological Services for MHEWS	
15:15 – 15:35	Break
15:35 – 17:30	• Session 11: From Operational Workflows to Pilot Implementation Roadmaps Dr. Caroline Staub, Ms. Emily DuBois ○ Guest Speaker: Dr. Abubakr Salih Babiker (WMO) ○ Apply lessons learned to refine each country pilot and define integration, deployment, and data-ingestion needs. Develop and present country-specific pilot implementation roadmaps ○ Discuss shared lessons and opportunities for continued collaboration

Day 4 September 17	
Time	Session
08:00 – 09:00	• Welcome • Sign in • Coffee

09:00 – 10:40	• Session 12: Institutional Considerations for Long-Term Sustainability and Scalability Mr. Washington Otieno ○ Ownership and governance ○ Organizational roles and responsibilities ○ Capacity development
10:40 – 11:00	Break
11:00 – 12:30	• Session 13: Institutional Considerations for Long-Term Sustainability and Scalability (<i>contd.</i>) Mr. Washington Otieno
12:30 – 13:45	Lunch
13:45 – 15:15	• Key takeaways • Next steps • Post-workshop evaluation • Concluding remarks Mr. Washington Otieno, WMO Regional Office for Africa, Mr. Habtemariam Tilahun, Hydrology and Basin Information and Dr. Fetene Teshome, Ethiopian Meteorological Institute (EMI)